

BCL1223 Database Fundamentals — Online Final Examination (May 2026)

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QUESTION 1

(a) Creation Script for All 3 Tables (15 Marks)

```
CREATE TABLE SALESPERSON (  
    SalespersonNo    CHAR(4)          NOT NULL,  
    SalespersonName  VARCHAR2(15)     NOT NULL,  
    Commission       NUMBER(5,2)      DEFAULT 0.00,  
    hireDate         DATE              NOT NULL,  
    CONSTRAINT pk_salesperson PRIMARY KEY (SalespersonNo)  
);  
  
CREATE TABLE PRODUCT (  
    productNo        CHAR(6)          NOT NULL,  
    productName      VARCHAR2(20)     NOT NULL,  
    unitPrice        NUMBER(8,2)      NOT NULL,  
    CONSTRAINT pk_product PRIMARY KEY (productNo)  
);  
  
CREATE TABLE SALES (  
    salesPersonNo    CHAR(4)          NOT NULL,  
    productNo        CHAR(6)          NOT NULL,  
    quantity         NUMBER(6)        NOT NULL,  
    CONSTRAINT pk_sales PRIMARY KEY (salesPersonNo, productNo),  
    CONSTRAINT fk_sales_salesperson FOREIGN KEY (salesPersonNo)  
        REFERENCES SALESPERSON(SalespersonNo),  
    CONSTRAINT fk_sales_product FOREIGN KEY (productNo)  
        REFERENCES PRODUCT(productNo)  
);
```

(b) Insert Script for 2 Records per Table (10 Marks)

```
INSERT INTO SALESPERSON (SalespersonNo, SalespersonName, Commission, hireDate)  
VALUES ('S001', 'Alice Tan', 0.12, TO_DATE('2022-01-15', 'YYYY-MM-DD'));  
  
INSERT INTO SALESPERSON (SalespersonNo, SalespersonName, Commission, hireDate)  
VALUES ('S002', 'Bob Smith', 0.08, TO_DATE('2023-05-20', 'YYYY-MM-DD'));
```

QUESTION 2

(a) Display all salespersons sorted by hire date (3 Marks)

```
SELECT * FROM SALESPERSON ORDER BY hireDate ASC;
```

(b) Display all salespersons with commission > 10% (4 Marks)

```
SELECT * FROM SALESPERSON WHERE Commission > 0.10;
```

(c) Display all products' unit prices after 10% increase (4 Marks)

```
SELECT productNo, productName, unitPrice,  
       unitPrice * 1.10 AS NewUnitPrice  
FROM PRODUCT;
```

(d) Display salespersons, product number and quantity for products Wrench or Hammer (6 Marks)

```
SELECT sp.SalespersonNo, sp.SalespersonName, s.productNo, s.quantity  
FROM SALESPERSON sp  
JOIN SALES s ON sp.SalespersonNo = s.salespPersonNo  
JOIN PRODUCT p ON s.productNo = p.productNo  
WHERE LOWER(p.productName) IN ('wrench', 'hammer')  
ORDER BY sp.SalespersonNo ASC;
```

(e) Display salesperson number, name, product name and total for each salesperson sold for each product (8 Marks)

```
SELECT sp.SalespersonNo, sp.SalespersonName, p.productName,  
       SUM(s.quantity * p.unitPrice) AS TotalSalesAmount  
FROM SALESPERSON sp  
JOIN SALES s ON sp.SalespersonNo = s.salespPersonNo  
JOIN PRODUCT p ON s.productNo = p.productNo  
GROUP BY sp.SalespersonNo, sp.SalespersonName, p.productName;
```

QUESTION 3

(a) Complete ERD for Order Form (15 Marks)

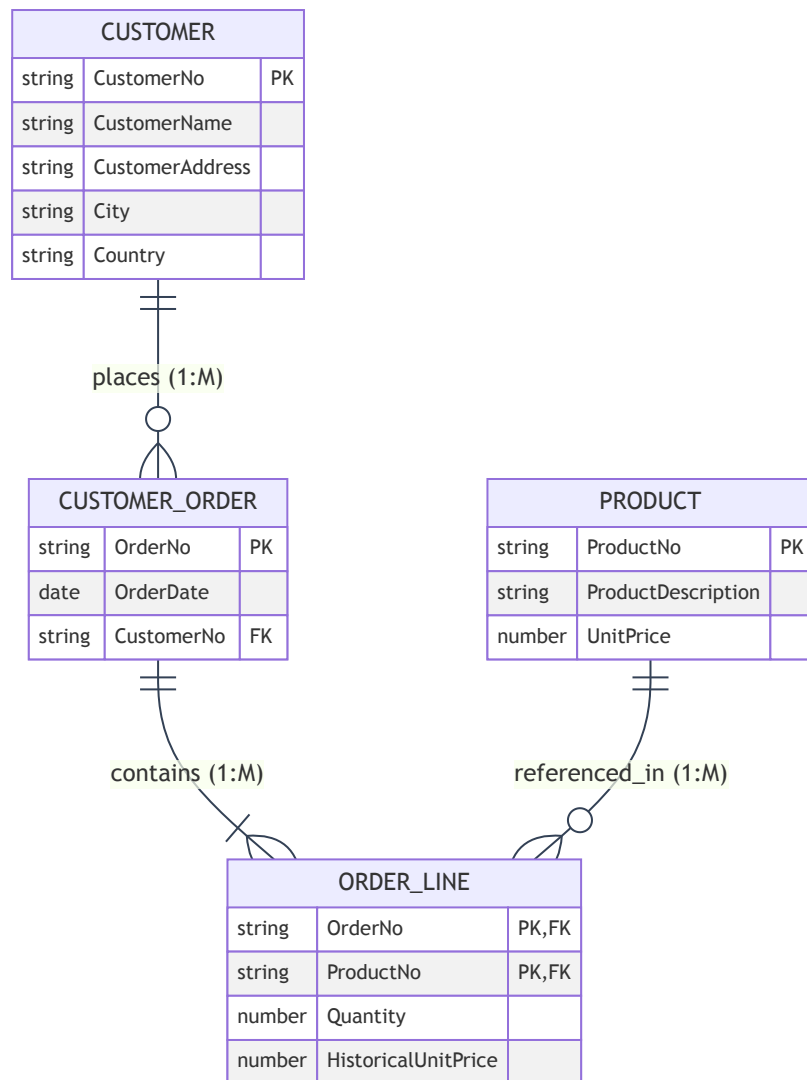
Entities & Attributes:

- CUSTOMER (CustomerNo, CustomerName, CustomerAddress, City, Country)
- CUSTOMER_ORDER (OrderNo, OrderDate, CustomerNo)
- PRODUCT (ProductNo, ProductDescription, UnitPrice)
- ORDER_LINE (OrderNo, ProductNo*, Quantity, HistoricalUnitPrice)

Cardinalities:

- CUSTOMER to CUSTOMER_ORDER : 1 to Many (1 : M)
- CUSTOMER_ORDER to ORDER_LINE : 1 to Many (1 : M)
- PRODUCT to ORDER_LINE : 1 to Many (1 : M)

ER Diagram:



(b) Oracle Creation Script for Table ORDER_LINE (10 Marks)

```
CREATE TABLE ORDER_LINE (
  OrderNo          VARCHAR2(10)  NOT NULL,
  ProductNo        VARCHAR2(10)  NOT NULL,
  Quantity         NUMBER(6)     NOT NULL,
  HistoricalUnitPrice NUMBER(8,2) NOT NULL,
  CONSTRAINT pk_order_line PRIMARY KEY (OrderNo, ProductNo),
  CONSTRAINT fk_line_order FOREIGN KEY (OrderNo)
    REFERENCES CUSTOMER_ORDER(OrderNo),
  CONSTRAINT fk_line_product FOREIGN KEY (ProductNo)
    REFERENCES PRODUCT(ProductNo)
);
```

QUESTION 4

(a) Normalization from UNF to 3NF (10 Marks)

- **UNF**: PRODUCTION_REQUEST (PR_No, Supplier_No, Supplier_Name, Supplier_Contact_Info, Order_Date, Signature, Expected_Delivery_Date, {Supplier_Item_No, Item_Description, Unit_Size, Unit_Price, Quantity,

Sub_Total})

- **1NF:** PRODUCTION_REQUEST (PR_No, Supplier_Item_No, Supplier_No, Supplier_Name, Supplier_Contact_Info, Order_Date, Signature, Expected_Delivery_Date, Item_Description, Unit_Size, Unit_Price, Quantity)
- **2NF:**
 - PR_HEADER (PR_No, Supplier_No, Supplier_Name, Supplier_Contact_Info, Order_Date, Signature, Expected_Delivery_Date)
 - ITEM (Supplier_Item_No, Item_Description, Unit_Size, Unit_Price)
 - PR_LINE_ITEM (PR_No, Supplier_Item_No, Quantity)
- **3NF:**
 - SUPPLIER (Supplier_No, Supplier_Name, Supplier_Contact_Info)
 - PRODUCTION_REQUEST (PR_No, Order_Date, Signature, Expected_Delivery_Date, Supplier_No*)
 - ITEM (Supplier_Item_No, Item_Description, Unit_Size, Unit_Price)
 - PR_LINE_ITEM (PR_No, Supplier_Item_No, Quantity)

(b) Database Anomalies & Examples (15 Marks)

1. **Insertion Anomaly:** Inability to add a record without inserting dummy/NULL values for missing primary key attributes (e.g., adding a course before any student enrolls in a combined table).
2. **Deletion Anomaly:** Unintentional destruction of secondary data when deleting a primary record (e.g., deleting the last staff member in a department permanently erases department information).
3. **Modification Anomaly:** Data inconsistency caused when updating redundant attributes across multiple rows fails to alter every instance (e.g., updating a customer address in 5 out of 10 order records).